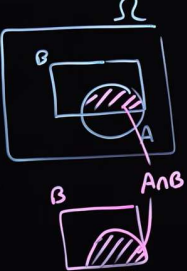


Conditional Probability

A, B are two events (outcomes) of a random experiment

Prob of A given B

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$


Ex. A = first die = 3
B = second die = 5
C = sum dice = 6

$P(A|B) = P(A) = 1/6$
 $P(A|C) = ??$

A = spades
B = black
C = red

Test 1000 people for cancer

500 cancer	450 + 50 -
500 don't have cancer	100 + 400 -

$P(A|B) = .9$

?? $P(B|A)$??

Bayes' Theorem Inference

Think about what the probability of **A given B** means: the probability of finding **A after B has happened**.

Here, don't get confused that we are finding the probability of A alone. We are finding the probability of **A given that B has already happened**.

So, if we have two dependent events, A and B, which have a common area representing **A and B**, then $P(A \cap B)$ is the probability that both A and B happen.

The probability of A after B has happened is the probability of **A and B**, divided by $P(B)$, which represents the effect of B having already happened.

So, this is how we get the formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Problem:

cancer patients – 500 have cancer and other 500 do not have cancer, but the machine says 450 positive, 50 negative instead of 500+. So, 50 of the results are wrong and the machine says 100+ even they don't have cancer,

so, let says $p(a)$ is having cancer, which is $500/1000 = 50\%$

$p(a)$ is device saying positive = $550/1000 = 55\%$

given that if b already happened which says 500 have cancer, now what is $\text{prob}(a | b) = 450/500$

So, for $p(b | a)$ = if a already happened that 550 people are tested +ve, but how many people actually have cancer? = $450/550 = 9/11$